

Interaction Design for Social Impact

CM2010 Sociocultural Perspectives to Innovative Technologies

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2026-04-22

Agenda

Session 1 (13.15-14.00)

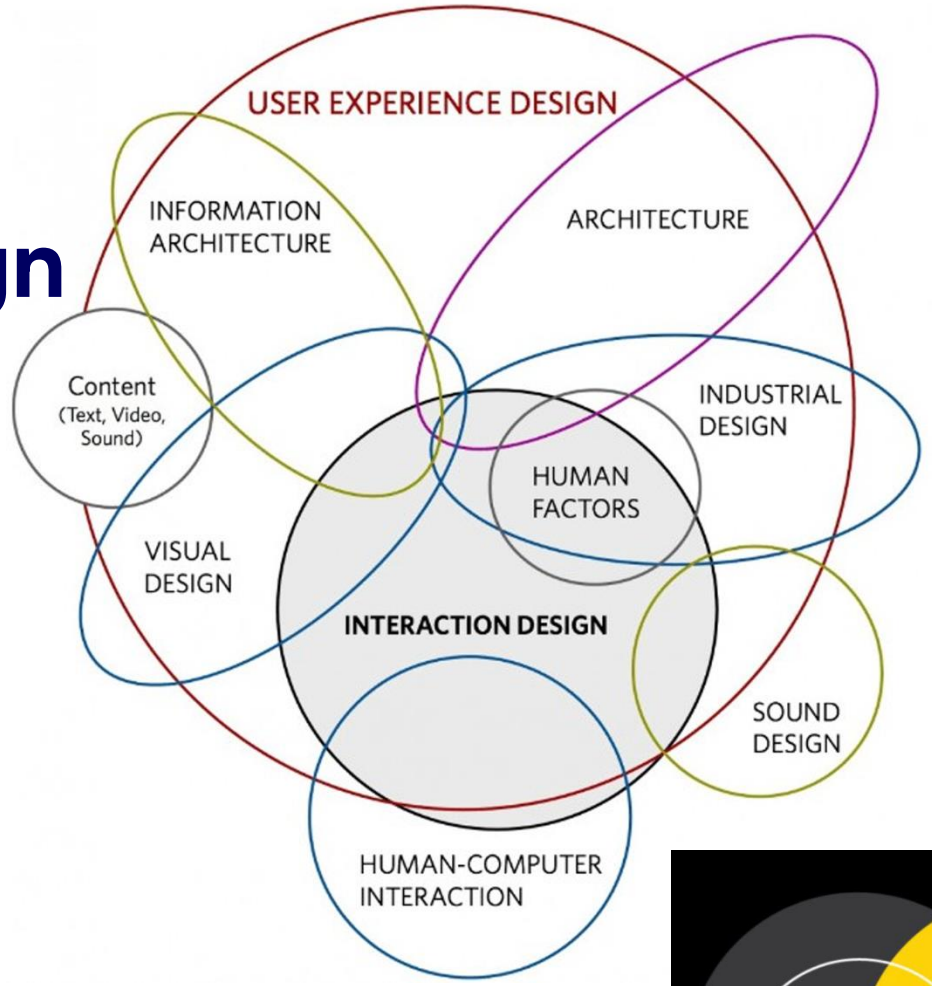
- What is Interaction Design
- Design examples with societal concerns
- Design fiction - induce social impact in your design process

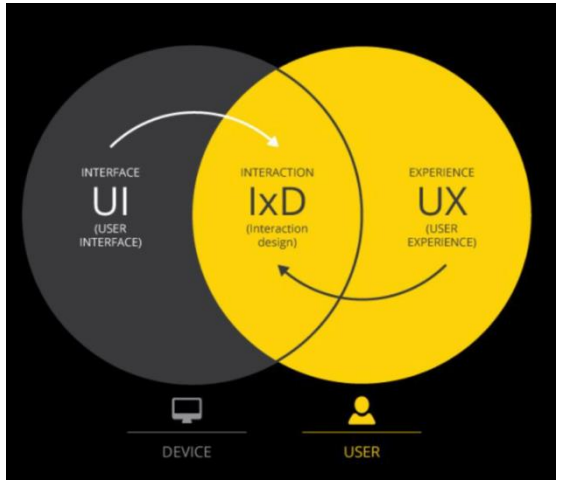
Session 2 (14.15-15.00)

Design for therapy and rehabilitation
Movement-based Design
Somaesthetic Design

What is Interaction Design

- A field focused on designing user experiences (UX) with interactive systems and products
- Examines how users interact with technology to create meaningful, context-specific experiences
- Aims to meet users' needs and expectations in various environments





Interaction Design's Multi-Disciplinary Nature

Combines knowledge from multiple fields, including:

- Graphic design
- Software engineering
- Human-computer interaction
- Cognitive psychology
- Anthropology
- Ergonomics
- Game design
- Product design

Goals of Interaction Design

- Create engaging and intuitive user experiences

- Ensure systems are easy to use, efficient, and enjoyable

- Applies to a wide range of digital systems:
  - Websites and mobile apps
  - IoT devices (e.g., smart appliances, autonomous vehicles)
  - AR/VR…

What is Social Impact in Design?

“A significant, positive change that addresses a pressing social challenge.” — Michigan Ross Center for Social Impact

“Social impact can be defined as the net effect of an activity on a community and the well-being of individuals and families.” — Centre for Social Impact (CSI)

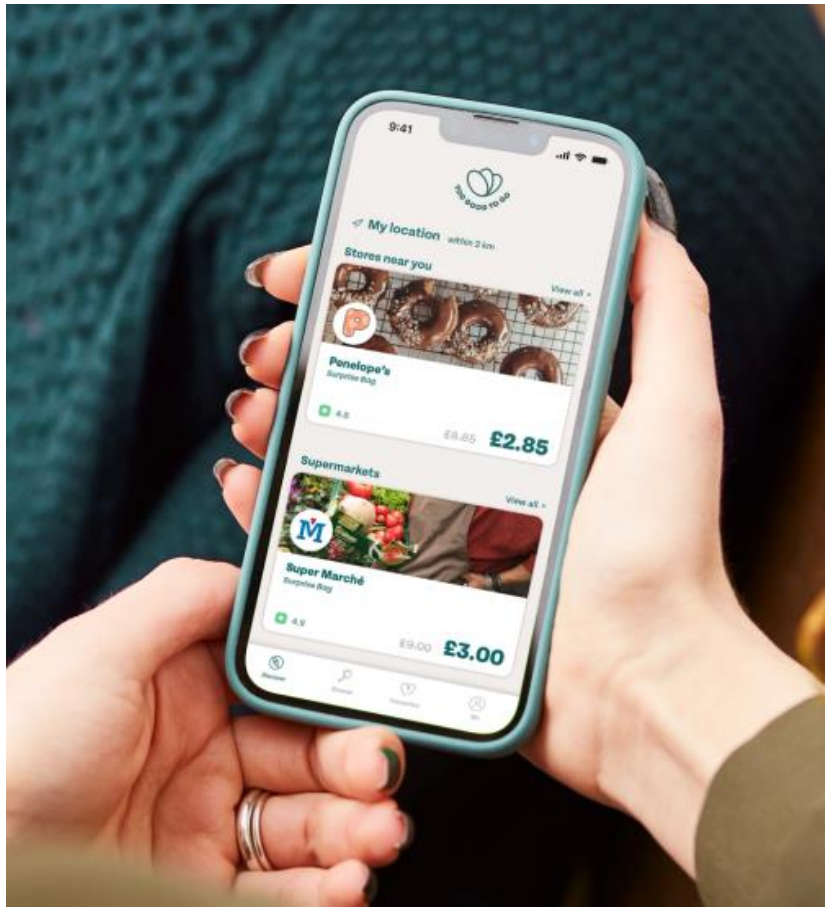
“Social impact applies to a broad spectrum of contexts. To designers, it is about the impact of products or services on individuals and groups of people... We think about balancing the needs of the individual with the needs of the overall community. On every design project, we take into account social, environmental, and economic impacts.“

— Innovation Design Engineering Organization

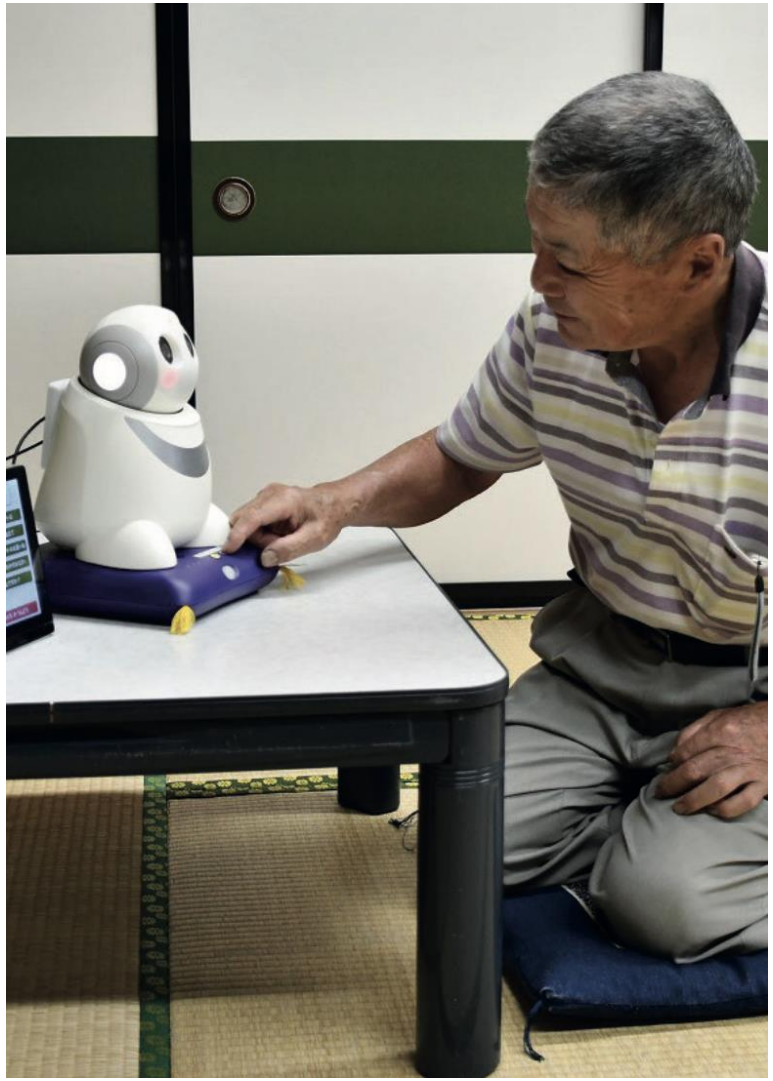


Inclusive space

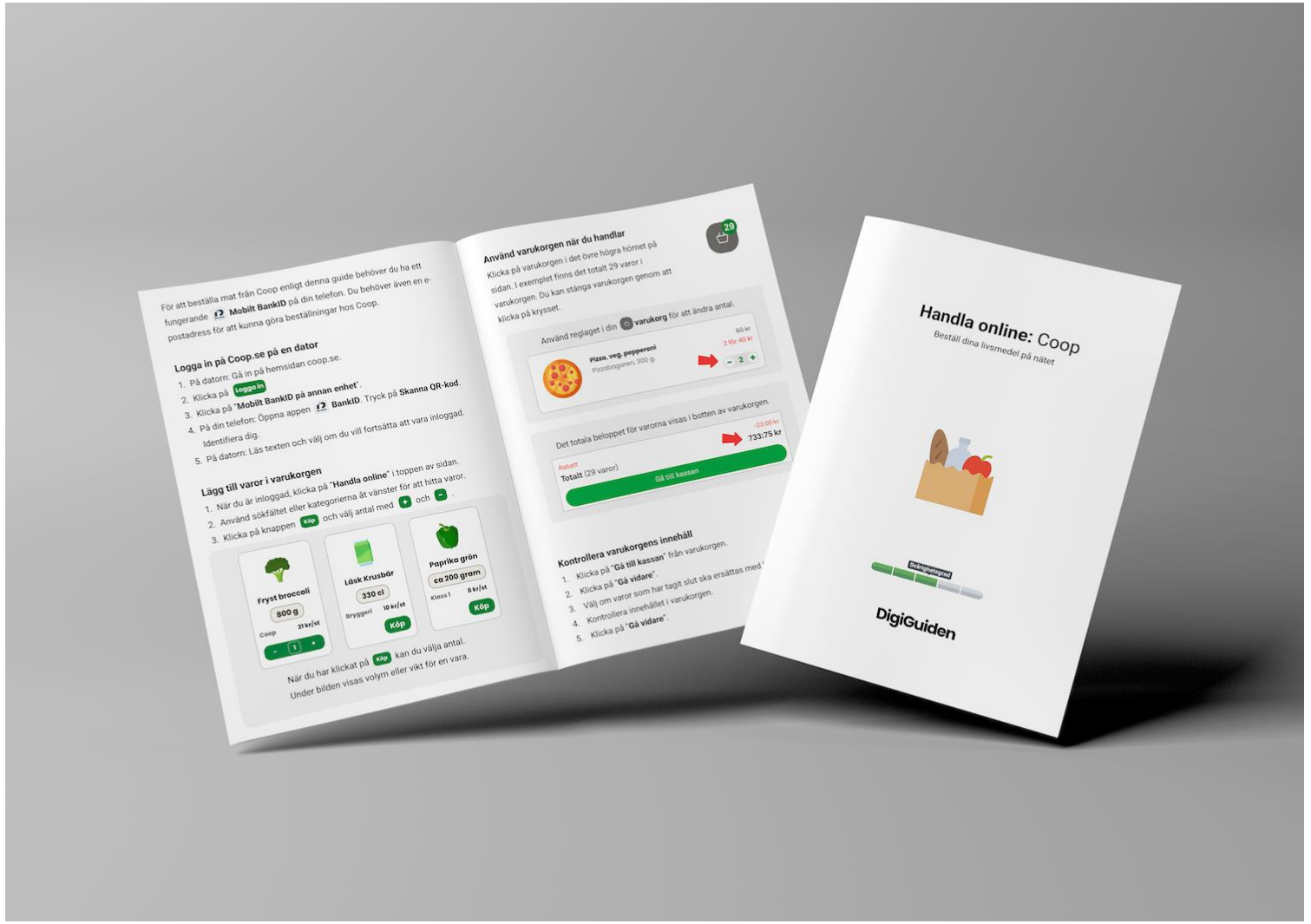
App (Too Good to Go)







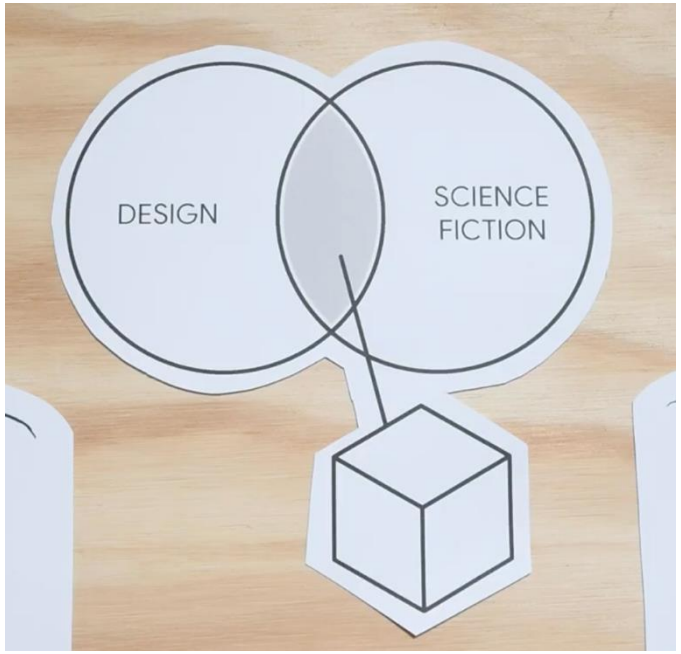
PaPeRo communication robot (2018)

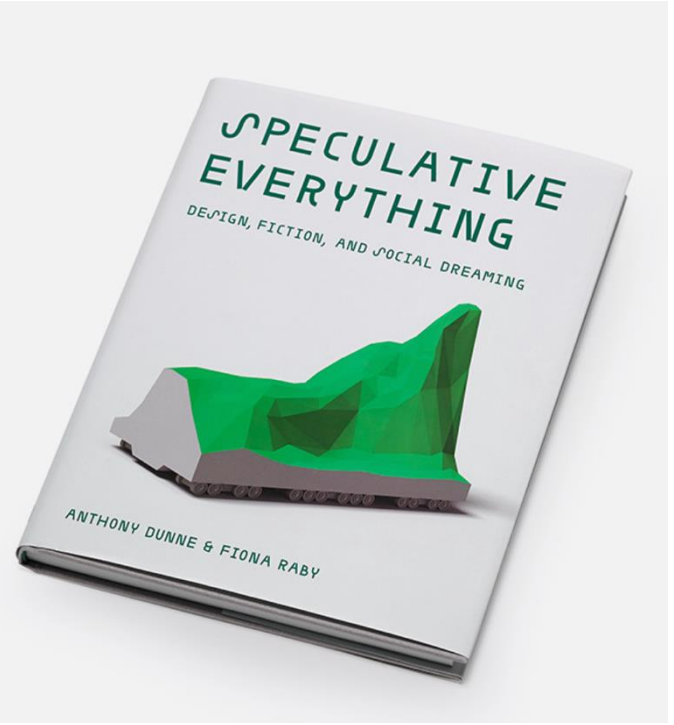


Digiguiden (2025)

Design Fiction

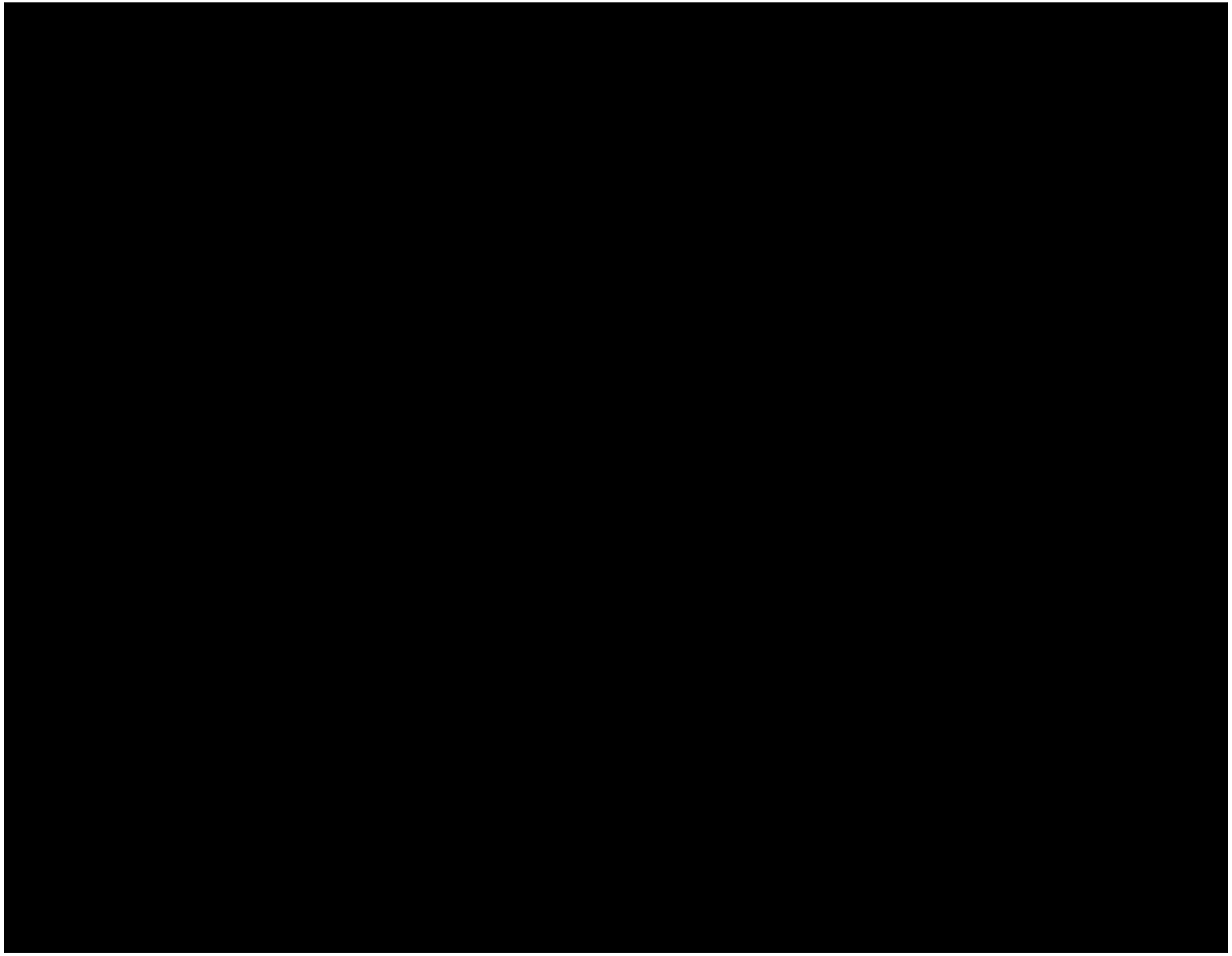
- ***Critical Design*** *uses* ***speculative design proposals*** *to* ***challenge*** *narrow assumptions, preconceptions and givens about the role products play in everyday life.*
- Design Fiction: The practice of creating tangible and evocative prototypes from possible near futures, to help discover and represent the consequences of decision making.





Speculative Everything - Design, Fiction, and Social Dreaming (2013)
https://www.youtube.com/watch?v=t_UT78JOauM

- It **visualizes “friction”** , moments where technology feels uncomfortable, invasive, exclusionary, or ethically problematic.
- To expose design flaws or blind spots (e.g., around accessibility, surveillance, bias).
- To **provoke discussion** about ethics, user agency, or unintended consequences.
- To **empathise with marginalised experiences** or challenge dominant norms.
- To **question assumptions** behind “smooth” or “efficient” interaction.



https://vimeo.com/96724979





https://dl.acm.org/doi/10.1145/2957276.2957298

The IKEA Catalogue: Design Fiction in Academic and Industrial Collaborations

Barry Brown et al. (2016)

Group Activity

- **Discussion - 15 mins**: reflect on the technology/interaction in the next design fiction video: https://shorturl.at/t1ce1



i. What are some challenges or frustrations the user face?
ii. What matters most to the user in situations like this?
iii. How do the user feel when using technology in this context?
iv. Who else is affected by this experience?





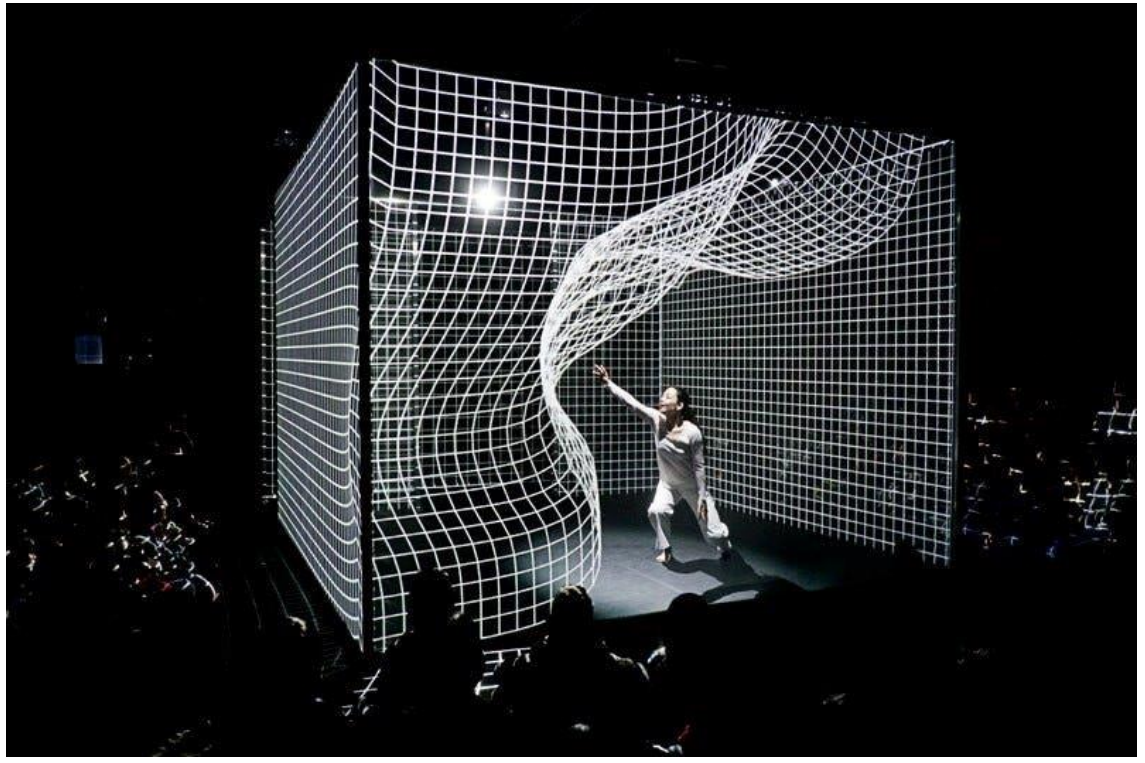
Interaction Design Beyond Screen

The “somatic turn” in HCI:

- shifts attention from the screen and cognition to the felt, lived body, how technology is experienced from the inside.

- *Somatic approaches treat the body as the primary site of experience, not just an input device.*

- They draw on somatic practices, dance, performance and phenomenology to understand and design interactions as felt, sensory, and affective, not only functional.



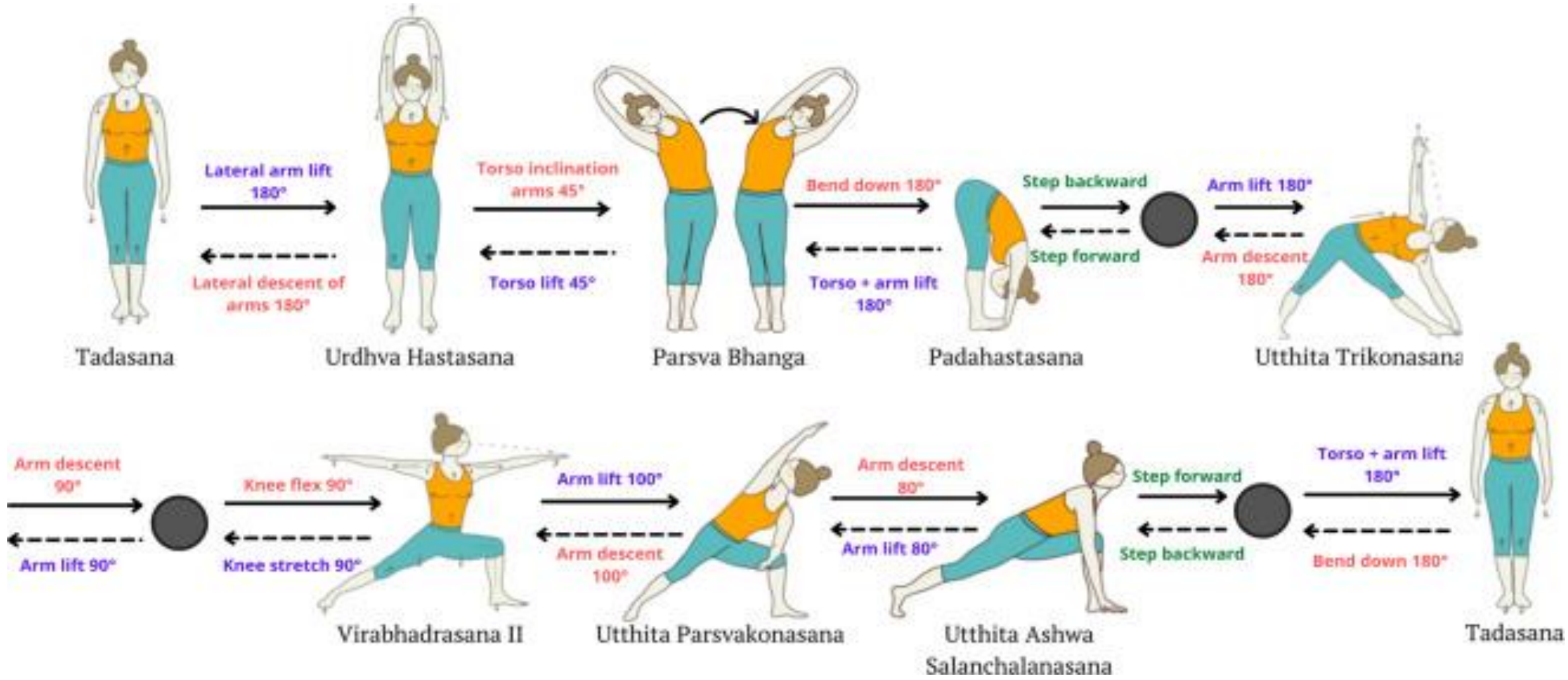


Lian Loke and Thecla Schiphorst. 2018. The somatic turn in human-computer interaction.

Why It Matters

- Aims for technologies that support richer, more meaningful experiences, not just efficiency
- Helps account for movement, emotion, and bodily agency in ubiquitous computing and human–computer integration
- In the context of rehabilitation, this approach moves beyond just monitoring physical metrics, aiming instead to foster deeper awareness, embodiment, and connection between the mind and body during recovery.

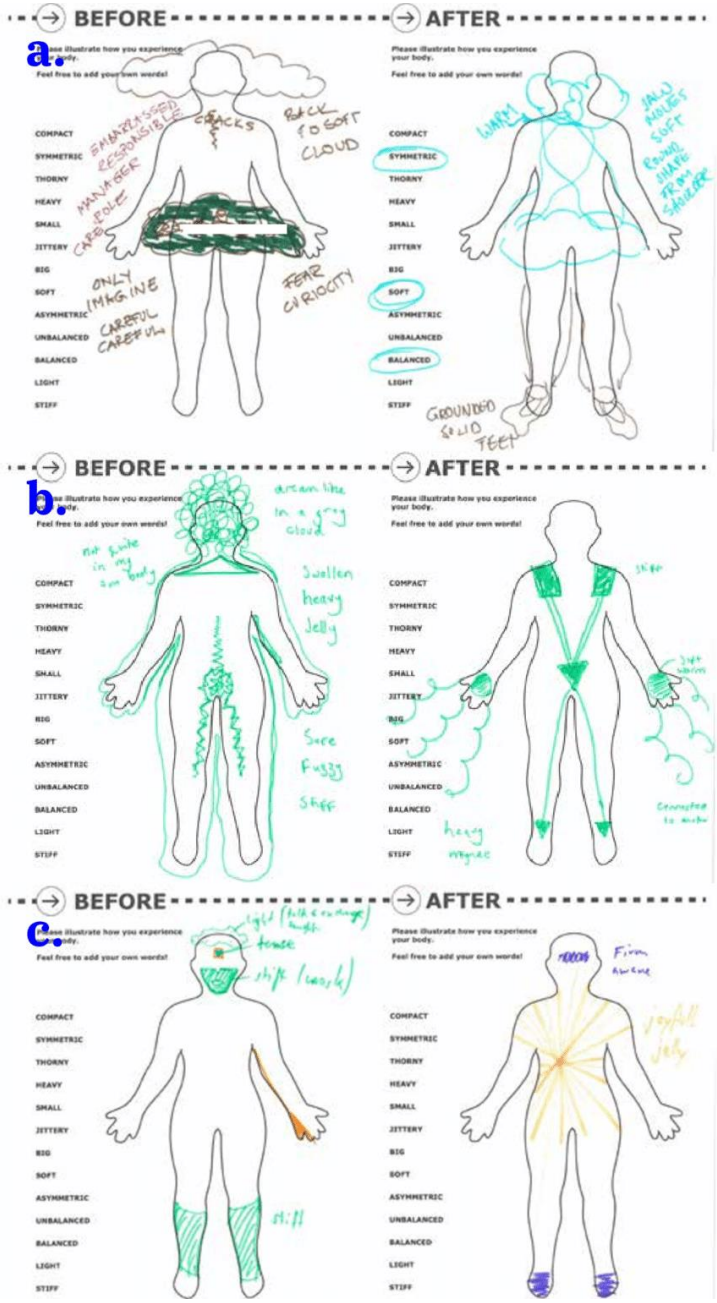




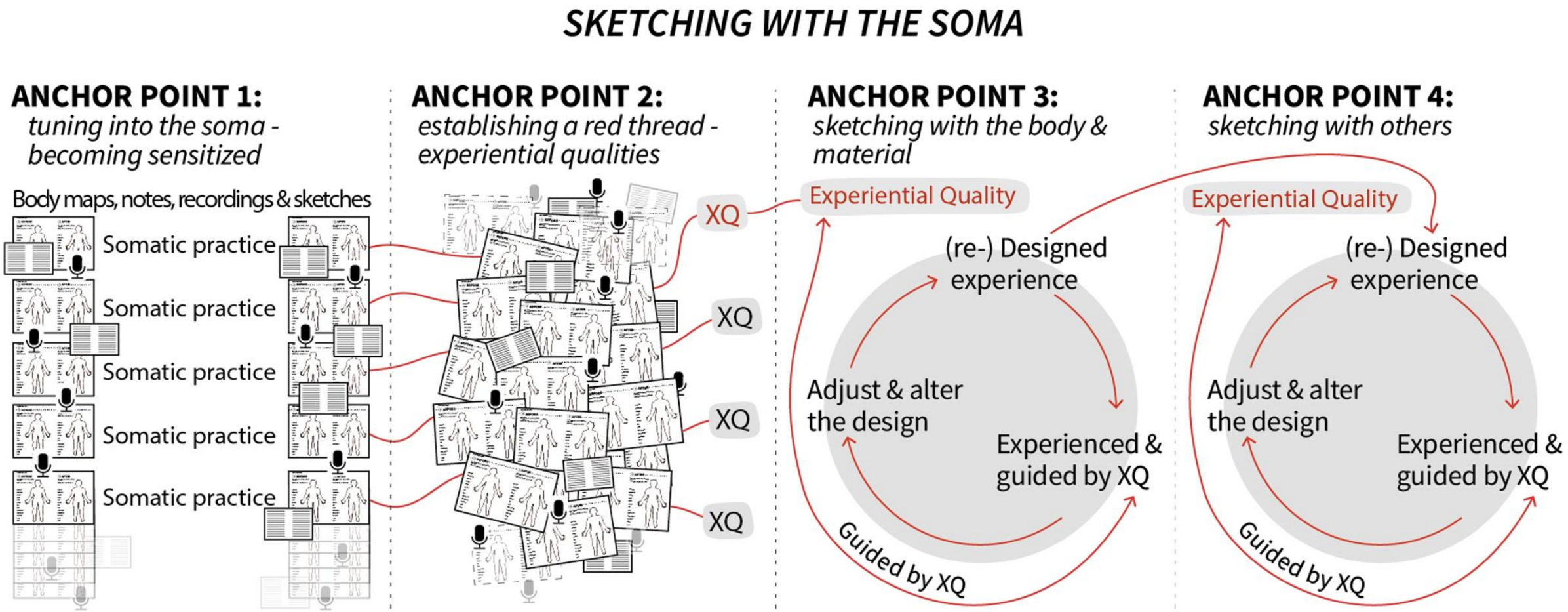
Zens: Designing and evaluating a movement sonification interface to alter body and flexibility perception in Yoga, https://doi.org/10.1016/j.ijhcs.2023.103084

Somaesthetic Design

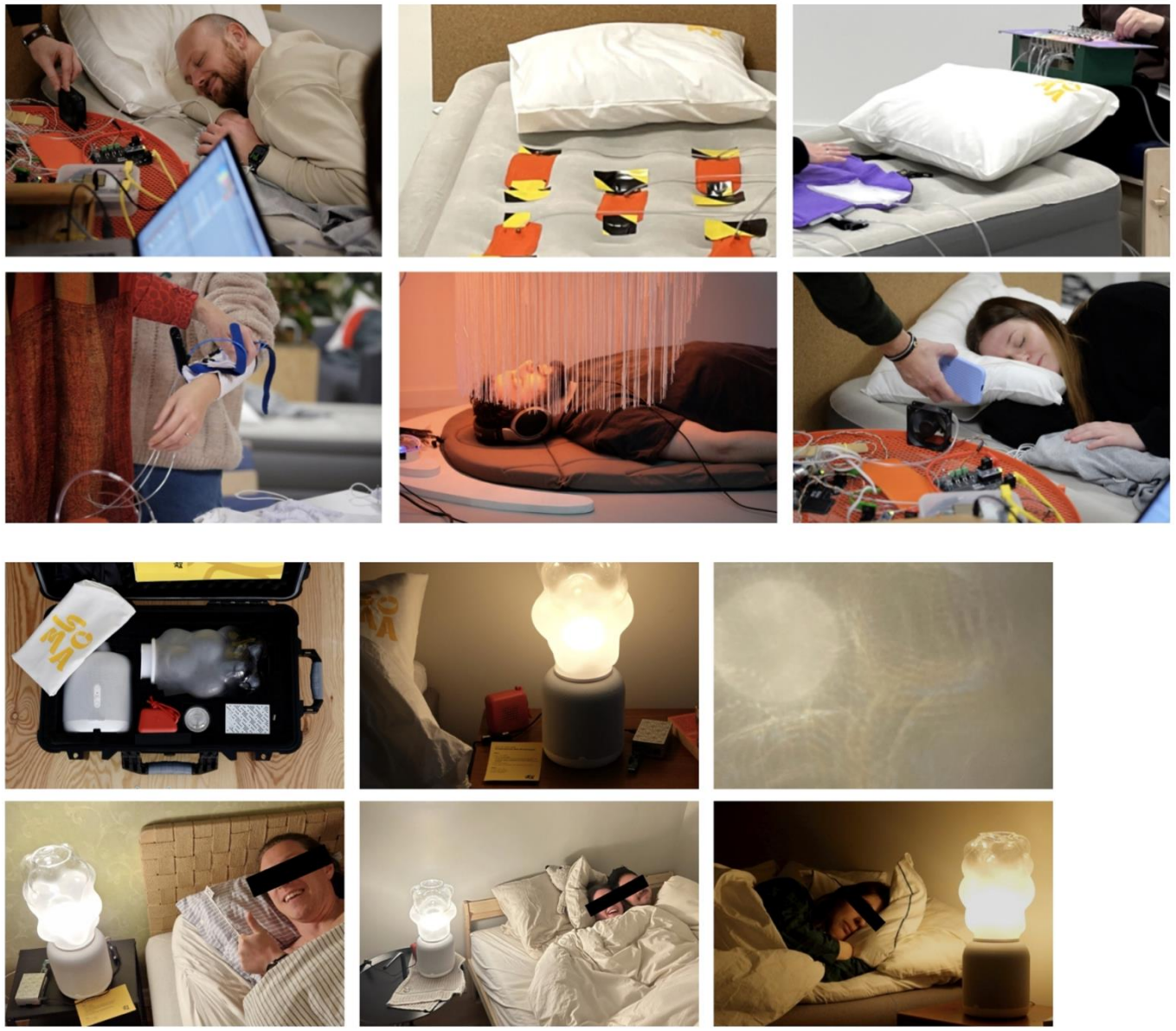
- **Rooted in philosophy**: Based on Richard Shusterman's *somaesthetics*, which emphasises the body (soma) as a site of sensory appreciation and self-cultivation.
- **Focus on bodily experience**
- **Mind-body integration**
- **Design as reflection and transformation**
- **Slow, sensory engagement**
- **Ethics of care and presence**
- **Applications**: Found in areas like health tech, wearable devices, rehabilitation, mindfulness tools, and interactive art.



Somaesthetic Design



Example:
The soma lamp



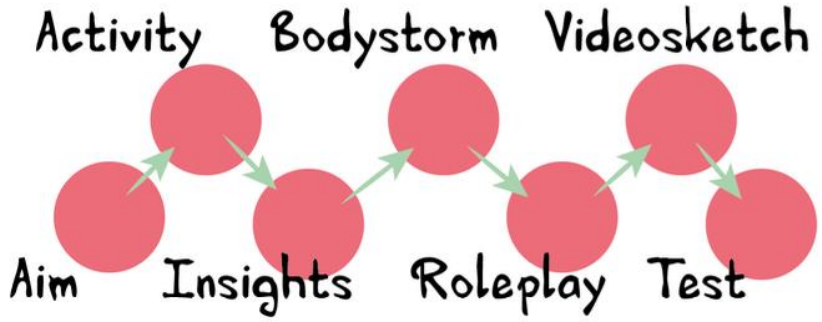
- **Designerly Ways of Knowing- Soma Design Meets Industrial Design Thinking** https://dl.acm.org/doi/fullHtml/10.1145/3686169.3686174

Comparing the core values

Soma Design (Academic Team)	Industrial Design (IKEA Team)
Deep first-person stance: Focuses on the designer's own bodily sensations.	**User-centered:** Focuses on empathy for the end-user and mass market.
Non-solutionist: Values to explore sensations in speculative ways.	**Solutionism:** Driven by a "value chain" from concept to sale.
Materiality: Uses experimental, expensive materials or prototypes (e.g., shape-changing robotics).	**Frugality:** Uses affordable, circular, and mass-producible materials.
Prioritise Feelings: Prioritises sensory engagement over verbal reasoning.	**Democratic Design:** Aims for "good quality at a low price" for everyone.

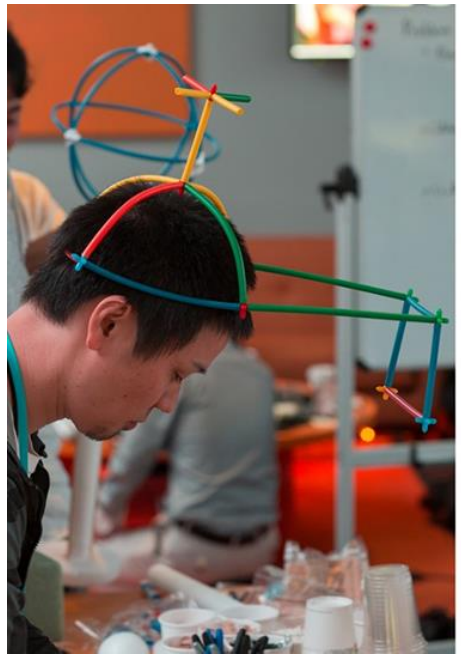
Movement-Based Design

- Centers the moving body: human motion as a central element in designing interactive systems and experiences.
- Beyond static interaction: how gestures, posture, rhythm, and full-body movement shape and inform interaction. It is Multisensory and spatial. And Supports alternative interactions
- Embodied knowledge: people think and feel through movement, drawing on insights from dance, somatics, and physical performance.
- Participatory and inclusive: Encourages co-design with users, especially those with different movement abilities or needs (e.g., elderly, disabled communities, dancers).
- Common contexts: Found in areas like interactive installations, wearables, sports tech, rehabilitation tools, and performative design.



Movement-Based Design





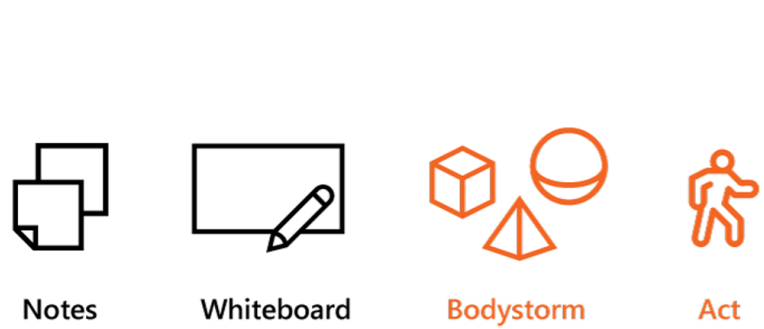


https://learn.microsoft.com/en-us/windows/mixed-reality/discover/case-study-expanding-the-design-process-for-mixed-reality





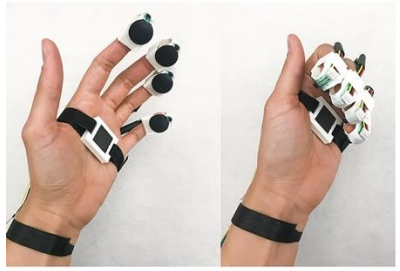



Notes
Whiteboard
Bodystorm
Act
Storyboard
OLC
One-line concept
Prototype
Asset creation

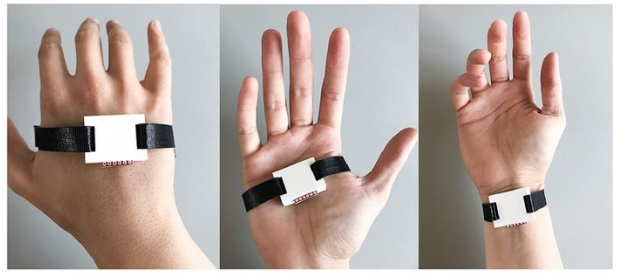


(a)

(c)



(b)



(d)



(a)

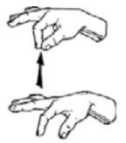


(b)



(c)

Felt sound's American sign language-inspired gesture-to-sound mapping

ASL gesture	Meaning	Sound engine	Detection
	Music	Low-frequency beating	Acceleration
	Show	Trigger drones	Magnetic sensing
	Poetry	Frequency change	Pressure sensing and acceleration
	Empty	Clear all sound engines	Magnetic and pressure sensing
	Discover	Add a new sound engine	Magnetic and pressure sensing

- Cavdir D, Wang G. Designing felt experiences with movement-based, wearable musical instruments: From inclusive practices toward participatory design. *Wearable Technologies*. 2022;3:e19. doi:10.1017/wtc.2022.15

Example: Age-Wear Project
Steps in the movement-based workshops





Research & Concept Introduction

Framing





Movement / Activity exploration

Somatic Exploration





Material-based Enactment of Wearable Scenarios

Body storming





Group Presentation and Reflection

Synthesis

Challenge

Movement Design Sheet *Feel free to use up all the space :)*

Group Name / Members:

Movement Scenario (e.g., balance training, rehab after surgery, dance for mobility.....):
leg training, balance

Target Group (age, mobility level, individual/group, etc.):
Seniors

Selected Materials (1–2): wrist bands / smart watch
balance mat

Placement of Wearables (1–2): handleder / wrists
fot matta / ~~balance mat~~ balance mat?

Primary Goals of Movement:
- ☒ Increase physical activity
- ☐ Improve motivation
- ☐ Support emotional connection with the body
- ☒ Enhance confidence and autonomy
- ☐ Other: _____

Design Considerations (e.g., comfort, safety, accessibility, flexibility, ….):
balancemat; make sure you are stable and don't trip/fall due to it

Resning från stol

Movement title:
Step-by-Step Movement Sketch & Description: Crising from chair







Example: Age-Wear Project
Steps in the movement-based workshops





Research & Concept Introduction





Movement / Activity exploration





Material-based Enactment of Wearable Scenarios

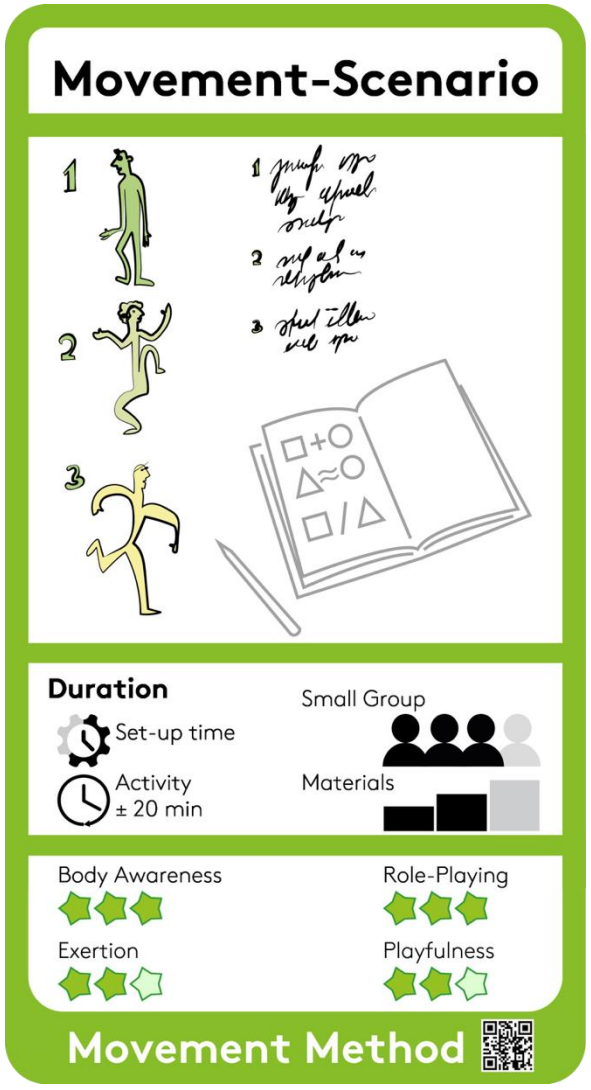


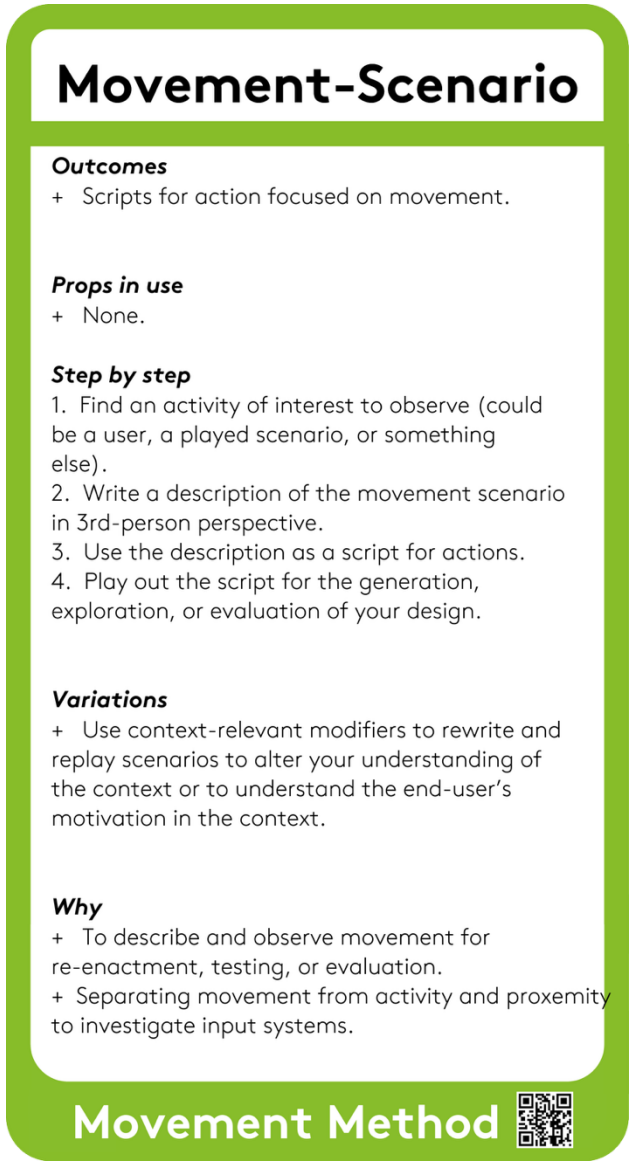


Group Presentation and Reflection

Movement-based design cards

https://mecamind.eu/







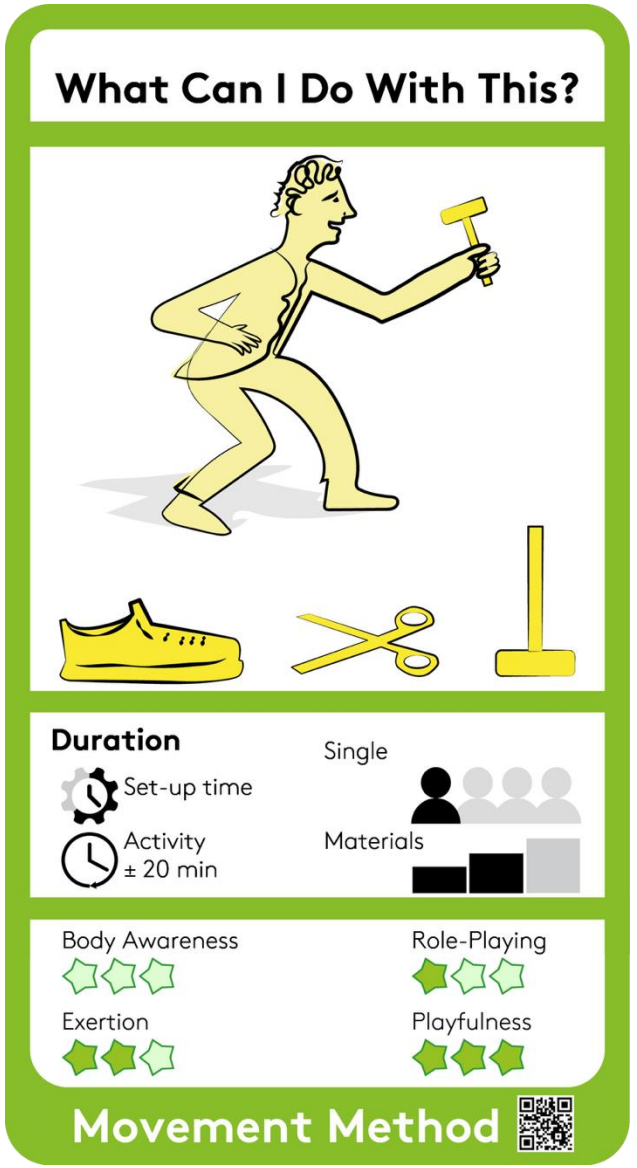
Movement / Activity exploration

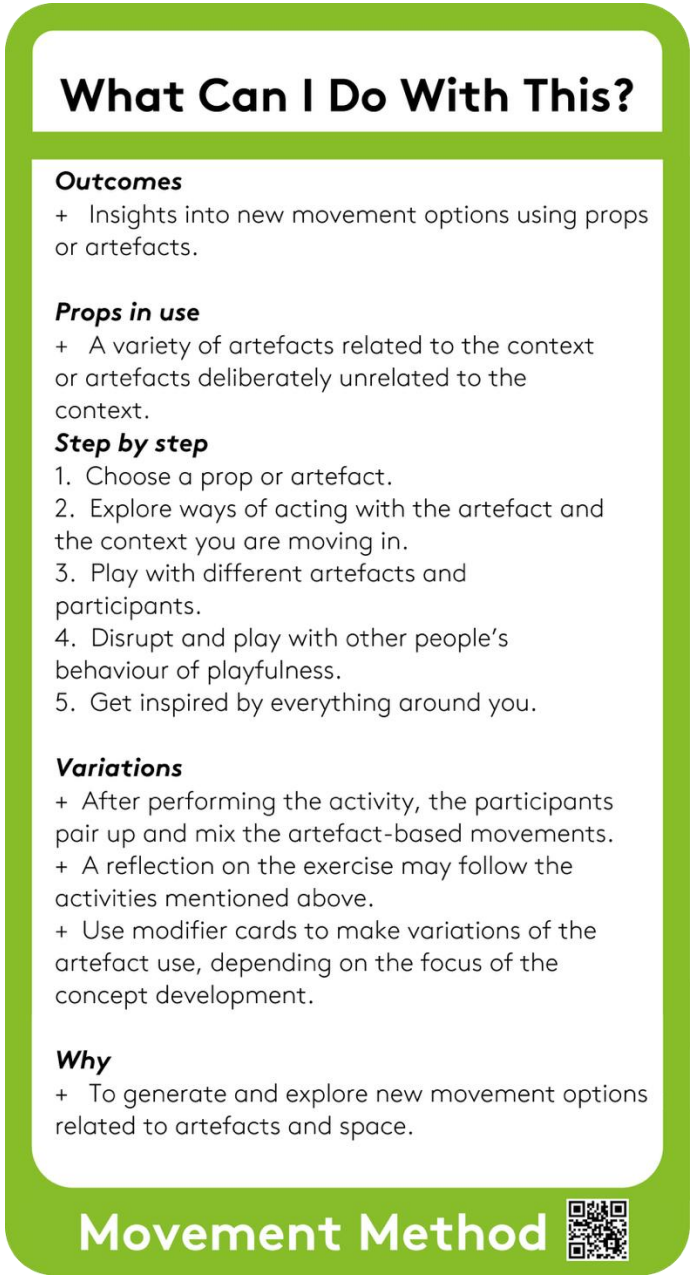
Movement-based design cards





Material-based Enactment of Wearable Scenarios





Group Activity – Movement Cards

Pick 1-2 workshop activities for movement method cards (https://mecamind.eu/ , green ones) for one of the flowing group:

– balance rehab at home after stroke
– social play for autistic children
– stress relief for overworked students

Annika Waern et al.
Moving with method: using cards in movement-based design, 2026
https://doi.org/10.1093/iwc/iwaf006

Consider:

1. The group size you want to work with?
2. The goal of the activity? – is it for you to design? to understand your users? Anything else?
3. Why do you pick these activities?

Document it here:
https://shorturl.at/t1ce1